

AI-driven Diversification for De Novo Enzyme Discovery

1 Introduction and Motivation

The AIZyme project aims to develop a computational pipeline for the *de novo* generation of enzymatic structures capable of reproducing catalytic functionality while exploring previously unknown regions of protein structure space. Existing protein engineering approaches are largely limited to modifying known enzymes, leaving most of the theoretical protein fold space unexplored. We hypothesize that modern generative AI models can be combined with structural prediction tools and machine learning-based analysis to systematically explore this manifold and identify candidates that preserve catalytic geometry while exhibiting structural diversity.

2 AI-driven Workflow for Diversified Enzyme Discovery

The proposed workflow consists of tightly coupled stages designed to generate, evaluate, and diversify candidate enzyme structures before experimental validation. The overall objective is to explore a broader region of the structural design space while preserving key catalytic motifs, increasing the probability of discovering novel enzyme scaffolds with improved or entirely new catalytic capabilities.

2.1 Generative Structural Exploration

Candidate backbones are generated using diffusion-based protein generative models [3], enforcing geometric constraints from reference enzymes. For each backbone, multiple sequences are predicted with `ProteinMPNN` [2]. These sequences are evaluated using `AlphaFold` [1], providing 3D coordinates and confidence metrics (pLDDT, PAE).

2.2 Structural Validation and Functional Preservation

Global similarity (RMSD) and catalytic site descriptors (geometry, pocket volume) ensure functional preservation. Candidates failing to reproduce key catalytic motifs are discarded.

2.3 Graph-based Structural Representation and Feature Extraction

Structures are converted to spatial graphs (residues = nodes; edges = proximity/interactions). Graph-based descriptors include centrality, density, radius, minimum dominating set size [10,11], and maximum quasi-cliques [12]. Meta-heuristic algorithms [13,14] compute these features efficiently, tuned using **Optuna** [16], executed in parallel on the VEGA HPC cluster.

Feature vectors $x_s \in \mathbb{R}^d$ capture multi-dimensional structural and topological properties for downstream analysis.

2.4 Diversity-aware Clustering

Unsupervised clustering (e.g., k -means, hierarchical clustering) partitions candidate structures [15]. Representative structures from each cluster are sampled to maintain diversity, while discarded structures are replaced via new generative cycles. This iterative loop ensures exploration of multiple structural regions.

2.5 Adaptive Diversification via AI-guided Feedback

Three AI strategies guide exploration:

1. **AlphaFold-based diversification:** Low pLDDT regions are targeted for structural variation.
2. **Graph Neural Network scoring:** Predicts stability and catalytic potential from residue interaction graphs.
3. **Active learning loop:** Experimental results refine predictive models and guide the next generation of candidates.

3 Research Questions and Objectives

Research Questions:

- RQ1: Can generative AI explore protein structures inaccessible via natural evolution?
- RQ2: How can structural diversity be quantified and actively enforced during enzyme design?
- RQ3: Can graph-based descriptors reveal structural families with catalytic potential?
- RQ4: Do AI-generated scaffolds preserve native catalytic geometry?
- RQ5: Can iterative AI–lab feedback progressively improve enzyme discovery?

Objectives:

- O1: Diffusion-based exploration of protein fold space.
- O2: Diversity-aware sequence and structure generation.
- O3: Graph-based feature extraction for stability and catalytic potential.
- O4: AI-guided clustering and candidate selection.
- O5: Closed-loop AI–experimental validation for progressive improvement.

4 Scientific Impact

AIzyme targets a transformative approach to enzyme discovery beyond naturally evolved proteins. Immediate impact includes enzymes for PET degradation and environmental applications. Long-term impact encompasses:

- Autonomous discovery of enzymes for green chemistry and sustainable synthesis,
- Design of biocatalysts for industrial biotechnology,
- Creation of novel metabolic pathways in synthetic biology,
- Establishment of a paradigm-shifting AI-driven enzyme discovery platform.

References

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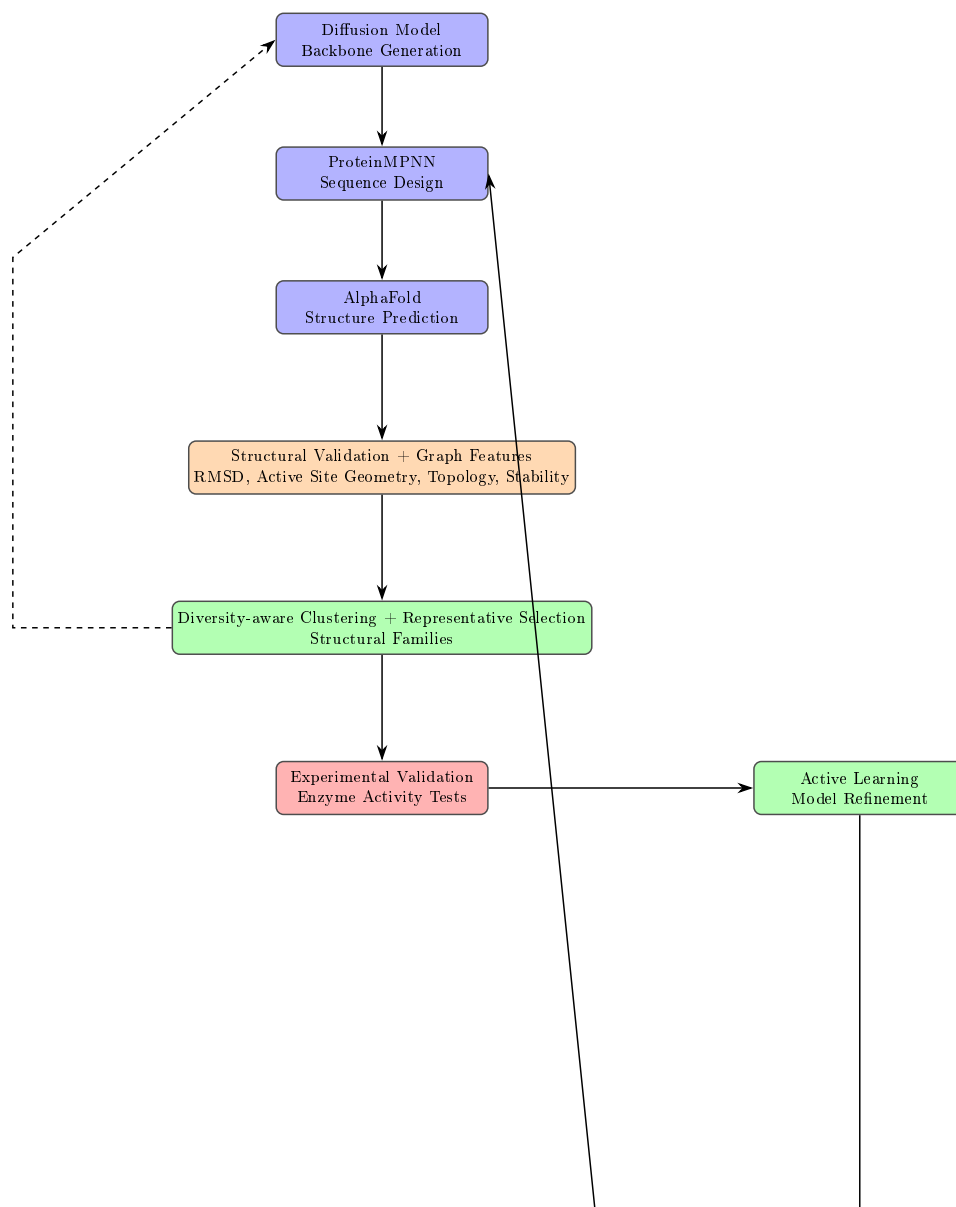


Fig. 1. Simplified AI-driven workflow for diversified de novo enzyme discovery. Core stages include generative backbone modeling, sequence design, structure prediction, structural validation with graph-based features, diversity-aware clustering, and experimental validation. Active learning feedback loops enable iterative improvement of candidate enzymes.